Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Fwd: Responses to papers 2, 5 and 6 revisions

Benjamin D. Summers <bsummers@adagiollc.com> Thu, Sep 24, 2026 at 1:17 AM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Below are Sara's comments on my responses to her reviews of Papers 2, 5, and 6...

---------- Forwarded message ---------
From: **ISMAIL-SUTTON, SARA R.** <sara.r.ismail-sutton@durham.ac.uk>
Date: Mon, Sep 21, 2026 at 2:45 PM
Subject: Responses to paper revisions
To: Benjamin D. Summers <bsummers@adagiollc.com>

Hi Ben,

Thank you for the detailed responses. I have gone through your explanations in relation to my original comments. As agreed, this response is limited to those explanations; I have not reviewed the revised manuscripts or independently checked the new proofs and calculations.

Memory Production under Coarse-Graining: A Balance Law for the Indivisibility Deficit

The changes you describe appear appropriately directed at my main concerns. In particular, the distinction between the exact deficit-level balance and the charge-level bounds, together with the clarification that Theorem 6.2 establishes monotonicity of the deficit norm rather than the charge, addresses the central interpretive issue in my review. The explicit comparison constants would strengthen the quantitative connection, subject to verification of the new argument.

I am happy with “two-time Chapman-Kolmogorov lumpability” in place of my suggested qualification of weak lumpability. The rename makes the distinction from process-level Markovianity clearer. The example you describe also directly illustrates why vanishing production need not eliminate higher-order dependence.

Your explanations concerning the law-weighted metric, the spaces and identifications in Lemma 2.3, and closure on the accessible subspace address the distinctions I was asking for. Similarly, retaining the flatness hypothesis and restricting the uniqueness claim to block-supported stochastic sections appropriately narrows the projection-related claims.

The remaining changes you describe, including the interpretation of production and severing, the limitations of the interference dial, and the status of the exact examples, are consistent with my recommendations. For the optimization discussion, the shared zero set should remain explicitly conditional on flat fine dynamics; two-sided comparison does not establish identical rankings or minimizers across cuts.

Memory Rank and Liftability of Transition Families Without a Generator: A Projection Theory for the Indivisibility Deficit

The changes you describe appear appropriately directed at my main concerns, particularly the distinction between closed-model approximability and exact positive-mass realization, the qualification of “memory rank”, and the separation of standard geometric facts from the model-specific contributions.

On finite termination, the uniform dimension bound and the closedness argument you outline provide the additional ingredient that my compactification comment identified as necessary. This is distinct from constructing a compactification at each fixed dimension. It therefore appears to address that concern at the level of the argument described, while leaving the exact-realization and boundary questions separate.

On the re-dressing class, I accept your point that these transformations have an established role in quasi-realization theory. My concern was specifically that they need not be invertible probability-preserving changes of coordinates, and that the trivialization conclusion should therefore remain relative to the declared class. The distinction you describe between this class, stochastic isomorphisms and conditioning-constrained transformations addresses that concern.

The joint-law obstruction also directly responds to the limitation I raised about pairwise marginal consistency. The distinction between a stronger detector and a stronger numerical lower bound is useful, as is retaining the qualification concerning positive marginals when discussing the converse.

Regarding Section 8, withdrawing the numerical study is a valid way to remove the claims and verification issues attached to it. My objection was to the strength of the conclusions drawn from the optimizer outputs, rather than to numerical illustrations themselves; cautiously presented feasible upper bounds could still have illustrative value. With those results and the claims relying on them removed, my request for their supporting archive no longer applies to the revised paper. The replacement exact examples would require their own verification, which is outside this response.

The remaining changes you describe, including the hidden-clock qualification, the revised first-order discussion, the narrower cohomological terminology and the expanded literature positioning, are consistent with the direction of my recommendations.

Koopman Spectral Theory for Indivisible Stochastic Processes

This follow-up concerns your responses only; I have not reviewed revision 4 or independently verified the revised proofs.

The corrections you describe to Proposition 2.11(c), the semigroup/evolution-family distinction, the generator scope, the intermediate-time bound and pointwise Kraus minimality appear appropriately directed at the issues raised. Removing the Michael selection-theorem argument also addresses that specific objection, although this does not independently verify the remaining completion argument, including the required behaviour along the diagonal. Restricting (C2) to the off-diagonal region addresses the incompatibility identified in my report.

On point 6, the hypothesis you quote is indeed imposed directly on the square roots. On that wording, my regularity objection does not apply, and I withdraw that comment. Thank you for drawing attention to the misreading.

Regarding Theorem 4.5(a), my concern was about the overall positioning of the contribution rather than the validity of the concentration statement. If the existing remark already identifies the standard Schmidt machinery, I am content with retaining that qualification. The useful distinction is between the structural consequence of how the deficit subspaces are defined and any additional dynamical information subsequently established.

Two qualifications from my review remain worth making explicit. First, the 1/3 bound should consistently be described as applying within the contraction class; column-stochasticity alone does not supply the required norm bound. Second, the revised augmentation passage should identify the particular operational restrictions and the Bell-type result supporting the exclusion. Referring generally to “operational constraints” would still leave the scope unclear.

The editorial changes and empirical caveats you describe are welcome. My request for a theorem-level comparison with non-Markovian transfer-operator, cocycle-defect and process-tensor approaches is broader than explaining the title or adding the collision-model reference; it concerns identifying precisely which results distinguish this framework from those approaches.

Best regards,

Dr Sara Ismail-Sutton